

CS

PORTFOLIO PROJECT

Customer Shopping Behavior Analysis

An end-to-end analytics project across Python, SQL and Power BI

3,900

Transactions

3

Tools Used

10

SQL Questions

2

Dashboard Pages

One dataset, three tools, one connected story

3,900 individual purchase transactions were traced through a complete analytics workflow: cleaned and feature-engineered in Python, interrogated with business questions in SQL, and brought to life in an interactive two-page Power BI dashboard.

The goal wasn't just to show charts — it was to move from raw transactional rows to decisions a retail team could actually act on.

DATASET SNAPSHOT

Rows	3,900
Original columns	18
Final columns	19 (after cleaning)
Missing data	37 nulls in review_rating
Customer segments	New / Returning / Loyal
Loyalty tiers	Occasional / Frequent / VIP

The same dataset, doing three different jobs



Cleaning decisions that protect data quality

1

Per-category median imputation

37 missing review ratings were filled using each product category's own median — not one dataset-wide median — keeping imputed values realistic.

2

Quantile-based age segmentation

`age_group` uses `pd.qcut()` to split customers into four equal-sized groups, so no single bucket dominates the analysis.

3

Redundancy check before dropping columns

Verified `discount_applied` and `promo_code_used` were identical on every row before removing the duplicate column.

Beyond SELECT / GROUP BY

Two of the ten business questions used more advanced SQL patterns:

WINDOW FUNCTION — Q8

Top 3 products per category

```
ROW_NUMBER() OVER (  
  PARTITION BY category  
  ORDER BY COUNT(customer_id) DESC  
) AS item_rank
```

Ranks products within each category independently — not just overall.

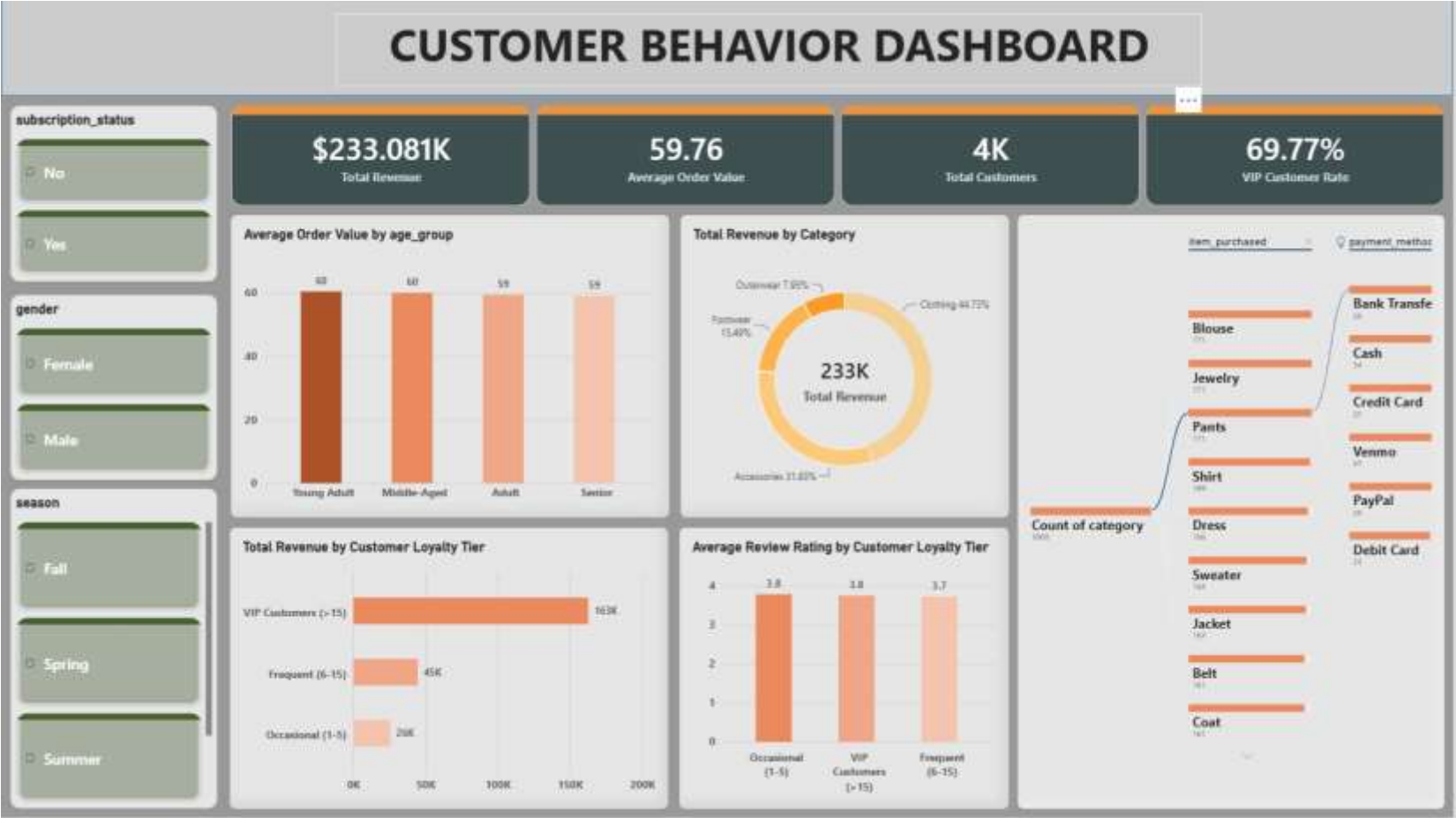
CTE — Q7

Segment customers by purchase history

```
WITH customer_type AS (  
  SELECT customer_id,  
         CASE WHEN previous_purchases = 1  
              THEN 'New' ... END AS segment  
  FROM customer_shopping_behavior  
)
```

Builds a temporary named result set to classify customers before aggregating.

Overview: the headline numbers



4 KPI cards, all driven by custom DAX measures

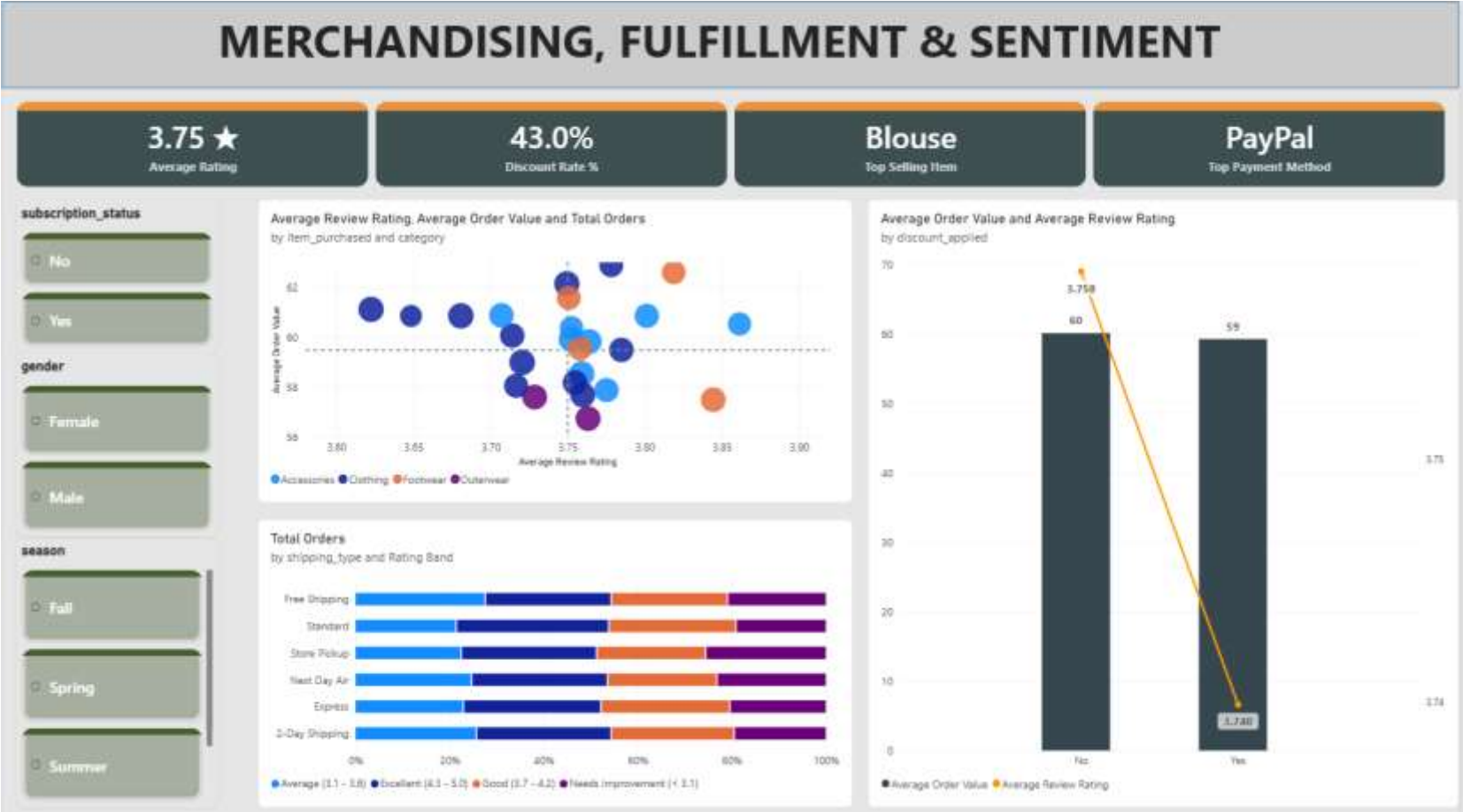


Customer Loyalty Tier: Occasional / Frequent / VIP



Decomposition tree drills category → item → payment method

Merchandising, Fulfillment & Sentiment



Scatter chart: rating vs. order value vs. volume



100% stacked bar: order ratings by shipping type



Combo chart: does discounting affect value & rating?

Integer division was silently truncating a KPI

BEFORE

	item_purchased text	discount_rate numeric
1	Hat	50.00
2	Sneakers	49.00
3	Coat	49.00
4	Sweater	48.00
5	Pants	47.00

SUM(...) / COUNT(*) performed integer division — every rate rounded down to a flat whole number.

AFTER

	item_purchased text	discount_rate numeric
1	Hat	50.00
2	Sneakers	49.66
3	Coat	49.07
4	Sweater	48.17
5	Pants	47.37

Changing “100 *” to “100.0 *” forces decimal math — precise rates like 49.66% now come through correctly.

SO WHAT?

Business Recommendations

1

Protect the loyal base first

3,116 of 3,900 customers are already Loyal — retention beats acquisition here.

2

Re-think the subscription program

Subscribers don't spend more per order; loyalty isn't converting to sign-ups.

3

Watch discount-dependent products

Hats, Sneakers, Coats, Sweaters, Pants each discount on ~half their sales.

4

Lead campaigns with top-rated items

Gloves, Sandals, Boots, Hats, Skirts — span 3 categories, not just one.

5

Target by behavior, not age

Revenue is flat across age groups — loyalty tier is the stronger lever.

TECH STACK

Built with



Python

pandas



PostgreSQL

SQLAlchemy



Power BI

DAX

Customer Shopping Behavior Analysis — Portfolio Project